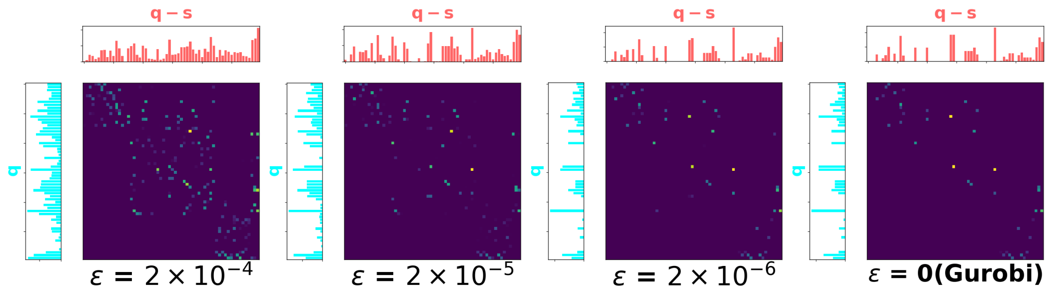

Figure 4: Visualization of the coupling and marginals by varying the regularization coefficient ϵ for EOFT. As ϵ decreases, the solutions of the coupling and marginals calculated by EOFT-Sinkhorn become increasingly sparse, approaching the exact solution.

coupling are positive ($\mathbf{q}^{(0)} > \tau$ and $\mathbf{q}^{(0)} - \mathbf{s} > \tau$) , making both $\mathbf{u}$ and $\mathbf{v}$ positive, thereby guaranteeing that $\mathbf{P}$ is positive. Regarding the calculation of $\mathbf{q}$, following Eq. 6, we can update it as:

$$\mathbf{q}^{(l+1)} = \frac{\mathbf{s}}{2} + \left((\mathbf{K}^{(l+1)}\mathbf{v}^{(l+1)}) \cdot ((\mathbf{K}^{\top})^{(l+1)}\mathbf{u}^{(l+1)}) + \frac{\mathbf{s}^2}{4} \right)^{\frac{1}{2}} - \mathbf{d} \tag{9}$$

In fact, for the iteration of $\mathbf{q}^{(l+1)}$, we would need to apply a truncation operation as done for $\mathbf{q}^{(0)}$ to ensure $\mathbf{q}^{(l+1)} \geq \tau$ and $\mathbf{q}^{(l+1)} - \mathbf{s} \geq \tau$. However, in practical iterations, we find that the computed result for $\mathbf{q}^{(l+1)}$ typically naturally satisfies these conditions. Furthermore, for the iteration of $\mathbf{K}$, we can utilize the constraints $\text{Diag}(\mathbf{K}) = \mathbf{d}$ in Eq. 5 to obtain:

$$\text{Diag}(\mathbf{K}^{(l+1)}) = \frac{\mathbf{d}}{\mathbf{u}^{(l+1)} \odot \mathbf{v}^{(l+1)}} \tag{10}$$

and the remaining matrix elements still maintain $\mathbf{K}_{ij}^{(l+1)} = e^{-\mathbf{D}_{ij}/\epsilon}$ for $i \neq j$. By iterating $l = 1, 2, \ldots$ until convergence, we can obtain the optimal solution $\mathbf{P}^*$ for entropic OFT. Similar to how the Sinkhorn algorithm for entropic OT often serves as an activation layer to ensure that the output of neural networks is a doubly stochastic matrix (Wang et al., 2019), our OFT-Sinkhorn algorithm can also serve as a layer to ensure that the output matrix satisfies flow balance constraints, which are typically enforced using loss functions in previous works (Bengio et al., 2023). Furthermore, since the matrix $\mathbf{P}^*$ exhibits backflow $\big($i.e., $\mathbf{P}_{ij}^* > 0$ and $\mathbf{P}_{ji}^* > 0$ for any node indices i and $j\big)$, we can perform a backflow removal operation $\mathbf{P}^* \leftarrow \max\{\mathbf{0}, \mathbf{P}^* - (\mathbf{P}^*)^{\top}\}$, which results in a $\mathbf{P}^*$ closer to the exact solution of OFT.

Global Convergence Then we give the convergence discussion. Following (Franklin & Lorenz, 1989), we adopt the Hilbert projective metric to prove our global convergence which is defined as:

$$d_{\mathcal{H}}\left(\mathbf{u}, \mathbf{u}'\right) \stackrel{\text{def.}}{=} \log \max_{i,j} \frac{\mathbf{u}_i \mathbf{u}_j'}{\mathbf{u}_j \mathbf{u}_i'}$$

Then we can get the theoretical results that our OFT-Sinkhorn algorithm has linear convergence and the proof are given in Appendix E.

Theorem 1. *The iterative scheme for OFT-Sinkhron algorithm has **linear** convergence. More precisely: one has* $\left(\mathbf{u}^{(l)}, \mathbf{v}^{(l)}, \mathbf{K}^{(l)}, \mathbf{q}^{(l)}\right) \rightarrow (\mathbf{u}^*, \mathbf{v}^*, \mathbf{K}^*, \mathbf{q}^*)$ *and*

$$d_{\mathcal{H}}\left(\mathbf{u}^{(\ell)}, \mathbf{u}^{\star}\right) = O(\lambda(\mathbf{K})^{2l}) \quad \textit{and} \quad d_{\mathcal{H}}\left(\mathbf{v}^{(\ell)}, \mathbf{v}^{\star}\right) = O(\lambda(\mathbf{K})^{2l}) \tag{11}$$

where

$$\lambda(\mathbf{K}) = \max_{l} \lambda(\mathbf{K}^{(l)}) = \sup\left\{ \frac{d_{\mathcal{H}}\left(\mathbf{K}^{(l)}\mathbf{v}, \mathbf{K}^{(l)})\mathbf{v}'\right)}{d_{\mathcal{H}}\left(\mathbf{v}, \mathbf{v}'\right)} \right\} \leq 1 \tag{12}$$

Additionally, we further discuss the numerical convergence of our OFT-Sinkhorn algorithm by varying the iterations in the experimental section.